

Choosing an antiderivative is not easy !

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When calculus students are asked to compute (by hand) an indefinite integral, they usually have access to an integral table which contains basic formulas they can use without justification and they need to use the classic techniques they have learned (usually: substitution, integration by parts, partial fraction expansion). But different integral tables, different textbooks, different computer algebra systems can provide different antiderivatives with different domains : see for instance [1], [2], [3] and [4]. The user may be familiar or not with complex valued functions. Some expressions such as $\operatorname{arctanh}(x)$ are real only for $-1 < x < 1$ even though it is an antiderivative of the expression $\frac{1}{1-x^2}$ everywhere on the real line except for $x = \pm 1$.

Let us recall that if f is a continuous function of the variable x over an interval $[a, b]$, then every function F such that $F'(x) = f(x)$ for $a < x < b$ is called an *antiderivative* of f over the interval (also called an *indefinite integral*). Because the derivative of a constant C is zero, we usually write

$$\int f(x) dx = F(x) + C.$$

In the case where an antiderivative of f can't be found in closed form, students learn that we can "construct" antiderivatives by using the formula

$$F(x) = \int_a^x f(t) dt \quad (a \leq x \leq b)$$

because this F is differentiable on $]a, b[$ with $F'(x) = f(x)$ for $a < x < b$.

The idea behind this talk is to start a reflection on the use of integral tables in mathematics courses in the era of computer algebra systems. Why? Despite the presence of more than 40 years of symbolic systems, it seems that they have had little influence on calculus textbooks. Our point is not to stop students computing integrals by hand. Not at all. When computing (simple) integrals by hand, students review their algebra (example: are the roots of the denominator real or non-real, should I use partial fraction or complete the square?), they need to use their derivatives formulas and for trig integrals, they need to use some basic trigonometric identities and all of these remain important. But we think that the following points should now be inserted into calculus courses curriculum – at least mentioned.

- It is important to recall that two antiderivatives of a function *that is continuous over an open interval* should differ by a constant but also to provide many simple examples. Some exercises should be done using pencil and paper and some more complicated should be using the symbolic/graphical capabilities of a CAS. One reason for this is the fact users of 2024 go to Wolfram Alpha and available CAS to compute and/or check integrals and the answers may vary from one system to another. Here are two examples where students should try to find the relationship between the different constants of integration.

$$\int 2 \sin(x) \cos(x) dx = \begin{cases} \sin^2(x) + C_1 \\ -\cos^2(x) + C_2 \\ -\frac{1}{2} \cos(2x) + C_3 \end{cases}$$

$$\int \frac{x^3}{(x+1)^5} dx = \begin{cases} -\frac{4x^3 + 6x^2 + 4x + 1}{4(x+1)^4} + C_4 \\ \frac{x^4}{4(x+1)^4} + C_5 \end{cases}$$

- Among the main motivations for including a specific antiderivative into an integral table is the desire of *symmetry* between formulas, wherever it is possible. Thanks to Rubi ([5]), this is achieved in many cases. For example, if a is a parameter, Rubi uses the following symmetric formulas :

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) \quad (a \neq 0)$$

$$\int \frac{1}{a^2 - x^2} dx = \frac{1}{a} \operatorname{arctanh}\left(\frac{x}{a}\right) \quad (a \neq 0)$$

For the last integral, *even when a is numeric*, Rubi still uses the arctanh formula (while other systems instead use logs).

- But if one goal is to have symmetry, the price to pay involves using antiderivatives that might be non-real outside an interval and, in certain cases, non-real at all ! Because complex numbers will cancel during the computation of a definite integral, this is not in fact a problem.
- Another motivation for including a specific antiderivative into an integral table is the importance to provide a continuous antiderivative : (at least) over the interval where the integrand is itself continuous. David Jeffrey was one of the first to publish about that, 30 years ago. The reference [6] contains many interesting papers.
- For a given integrand, students should be able to prove that a list of different expressions constitute valid antiderivatives. They must also be able to find the domain of each antiderivative. For example, consider again the expression $\frac{1}{a^2 - x^2}$ with $a \neq 0$. If we decide to use partial fraction expansion, then the antiderivative contains logs. And depending if the real or the complex branch is selected, absolute values may appear in the logs. Table 1 shows some antiderivatives calculus students must be able to deal with in today's CAS environment. Graphs that help to visualize the domain in the case $a = 1$ where performed using TI-Nspire Premium Teacher Software as Figure 1 shows: in Nspire, $\operatorname{arctanh}(x)$ displays as $\tanh^{-1}(x)$.

Table 1: Four different antiderivatives of $\frac{1}{a^2 - x^2}$ with their respective domain

Domain	Real antiderivative of $\frac{1}{a^2 - x^2}$
$-a < x < a$	$\operatorname{arctanh}\left(\frac{x}{a}\right) = \frac{1}{2a} \ln\left(\frac{a+x}{a-x}\right)$
$x > a$ or $x < -a$	$\frac{1}{2a} \ln\left(\frac{x+a}{x-a}\right) = -\frac{1}{2a} \ln\left(\frac{x-a}{x+a}\right)$
$x > a$	$\frac{1}{2a} \ln(x+a) - \frac{1}{2a} \ln(x-a)$
$x \neq \pm a$	$\frac{1}{2a} \ln\left(\left \frac{x+a}{x-a}\right \right)$

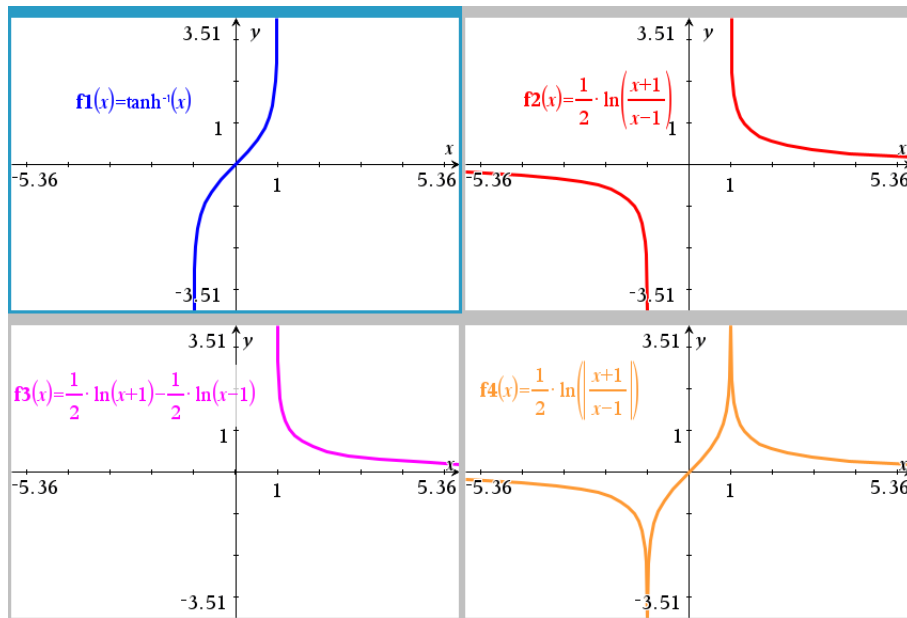


Figure 1: Different antiderivatives of the expression $\frac{1}{1-x^2}$ and their real graphs

When David Stoutemeyer and Albert started the development of **Derive**, they (probably) did not see the potential educational use of their software. But it soon became clear that **Derive** would become a fantastic tool for teaching. The ability of Rubi to show the steps during an integration process is going into the same direction.

Now, in order to start a reflection on the use of integral tables in mathematics courses in the era of computer algebra systems, we first need to start inserting hyperbolic functions into the curriculum (with their inverse functions as well). Then many usual integration formulas shown in textbooks can be replaced by some having a better look and that are easy to demonstrate. An example of this would be for students to show that

$$\int \sec(x)dx = \operatorname{arctanh}(\sin(x)) \quad \text{instead of using} \quad \int \sec(x)dx = \ln(|\sec(x) + \tan(x)|).$$

Let us be clear. We don't intend to introduce undergraduate students to more advanced subjects such as, for example, "Kahanian constant of integration" (see [7]). We only want to try to "update" the integral tables calculus students are using while being aware that several choices exist.

So one question is: if someone wants to select a single antiderivative, which one should be inserted into a table of integrals or chosen by a CAS? In order to answer this question, we will need additional time and this will be part of another talk that will best fit into the session *Integration and ODEs* (see [8]).

Keywords

Calculus textbooks, Tables of integrals, Computer Algebra Systems

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